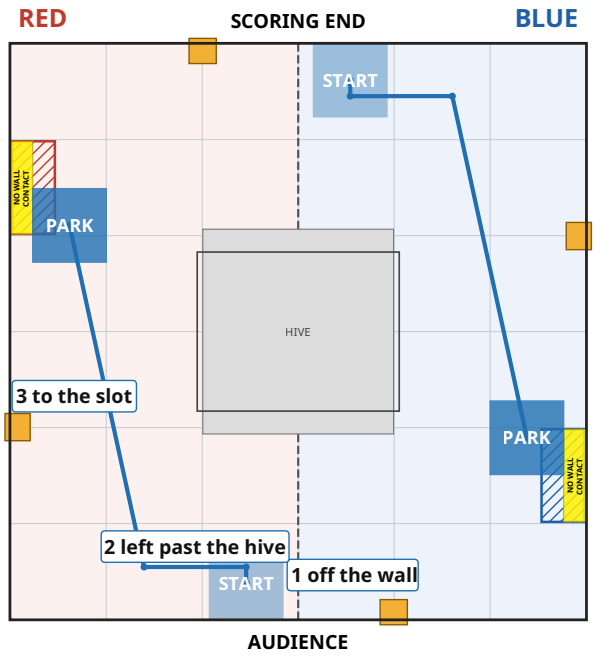
Odo ROBOT Shoot + Park

Use this when we shoot first. It is our main autonomous.

The robot shoots its 4 pollen, then drives to the parking slot on its own. It measures where it is as it drives (odometry), so it ends up in the same place every time.

What the robot does



Time	What happens
0 - 0.5 s	Drives gently **forward** into the wall in front of it, to line itself up straight. This is normal.
from 0.5 s	Launcher spins up: 3 s at most.
next 10 s	Shoots the pollen.
by about 19 s	Drives **backward** (launcher end first): (1) a little off the wall, (2) left past the hive, (3) to the **INNER** parking slot. Then it stops.

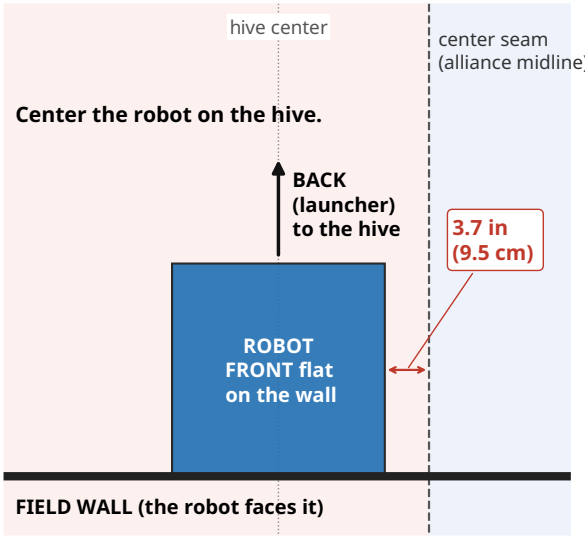
Points if it works: **LEAVE 3 + PARK 5**. Our partner parks in the **OUTER** slot.

If something blocks it on the way, it **stops where it is**. It never skips ahead to the next leg from the wrong spot, because that straight line could go through the hive.

Before the match

1. Agree with the partner. We shoot first. They shoot second (or don't shoot), stay out of our lane, and park in the OUTER slot. Show them the pit sheet.

2. Put the robot on the field.



- **Red:** at the **audience** wall. **Blue:** at the **scoring-end** wall.
- **Front** of the robot **flat against the wall**. The **back (the launcher) faces the hive**.
- **Center the robot on the hive.** Its edge is 3.7 in (9.5 cm) from the center seam.
- Exactly **4 pollen** loaded. Not touching a flower. Not in the loading zone.

3. On the Driver Hub, pick **Odo ROBOT Shoot + Park**. It's in the autonomous list, group *Pollen Odometry*.

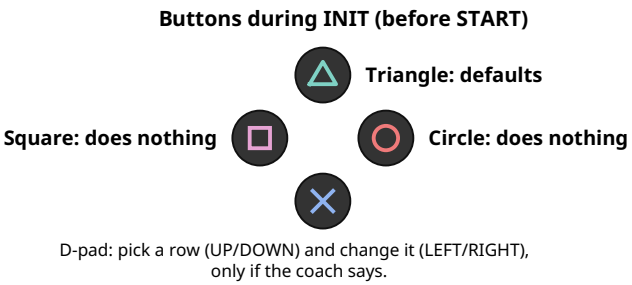
4. Press INIT. Don't touch the robot for 1 second: the odometry sensor is getting ready.

5. Press Triangle once. That puts every setting back to the defaults. Then make only the changes the coach asks for (usually none).

6. Check the screen:

Line	Should say
ROUTE (the first line)	`FULL PARK: shoot, then 3 legs to the parking slot`
Build	`pollen-hive-v4`
Mode	`ODOMETRY Shoot + Park`
Odometry x fwd / y left / heading	numbers near 0 (not `NOT VALID`)
Status	`READY - no movement until START`

7. Press START when the match starts.



- **Triangle:** back to the defaults. Always safe.
- **Square** and **Circle:** do nothing.
- **D-pad:** changes settings. Only if the coach says.

If something is wrong

You see	What it means	Do this
`BLOCKED: ... No goBILDA Pinpoint named "odometry"`	The odometry sensor isn't found	Use **EMERGENCY Timed Shoot + Park** instead (emergency timed page). Tell a programmer.
`Odometry ...: NOT VALID`	The odometry sensor isn't reading	Press STOP, then INIT again with the robot still. Still bad: use the emergency timed mode.
`READY FOR DEPARTURE ONLY - SHOOTER FAILED`	The launcher didn't start up	The robot will still drive and park, but won't shoot. Tell a programmer; check the launcher cables.
`NOT READY - over time budget`	Someone's settings take too long	Press Triangle.
ROUTE isn't `FULL PARK` (and the coach didn't ask)	Someone changed a leg	Press Triangle.
`PINPOINT FAULT`	A pod cable is loose	Check both pod cables. Still there: emergency timed mode.
Any other `BLOCKED`	Something isn't plugged in, or the wrong robot configuration is active	Tell a programmer.

After the match, tell the programmers

- **The match number** (the robot keeps a log; a programmer pulls it after the event).
- Did it shoot all 4? Did it park in the slot? Was it touching the side wall?
- **Photograph the screen before you INIT TeleOp**: it shows the last **Phase** and any **Warning**.